

SlabMaster™ Enterprise Platform

Executive Summary & Strategic Overview

Version: v1.0.0

Target Industry: Stone Fabrication, Quartz Manufacturing & High-Volume Countertop Installation

Deployment Surface: Cloud Multi-Tenant SaaS (Azure Static Web Apps + Container Apps) & Mobile Offline PWA

Updated: September 2026

1. Executive Summary

SlabMaster™ is a modern, enterprise-grade cloud fabrication and field execution platform purpose-built for stone fabricators, quartz manufacturers, commercial millwork contractors, and high-volume residential countertop installers.

Historically, the stone fabrication industry has been constrained by legacy client-server tools such as Moraware Systemize and CounterGo—technologies architected decades ago with antiquated desktop-bound UIs, brittle SOAP/.NET APIs, lack of native mobile capability, absence of true ERP integration, and no offline field support.

SlabMaster bridges the critical operational divide between corporate ERP systems (e.g., SAP S/4HANA, Microsoft Dynamics, NetSuite) and shop floor/field reality. It delivers an end-to-end operational platform spanning:

1. **Three-Tier Account/Community/Lot Hierarchy** tailored for national homebuilders and commercial developments.
2. **Automated Lead-Time Scheduling Engine** supporting plant-specific working days, holiday blackouts, and truck capacity limits.
3. **Live Slab & Remnant Inventory Control** with barcode/QR tracking, bin locations, lot allocations, and defect logging.
4. **Direct Materials Purchasing & Receiving** with automated PO generation, vendor tracking, and inventory reconciliation.
5. **Shop Floor Kiosk PWA** for touchscreen sawjet, CNC, and polish station tracking with one-tap milestone advances.
6. **Mobile Field Dispatch & Subcontractor Portal** with offline capability, GPS routing, digital photo uploads, and customer sign-off capture.

7. **Custom Table Matrix Forms & EAV Attribute Engine** allowing fabricators to capture arbitrary multi-row specifications without code changes.
8. **Modern RESTful ERP Two-Way Sync** featuring external ID resolution (`externalId`), Change Data Capture (CDC), and an automated Outbound Retry Queue with exponential backoff and dead-letter recovery.

2. Market Problem vs. The SlabMaster Solution

Operational Challenge	Legacy Solutions (Moraware / Spreadsheets)	The SlabMaster Advantage
ERP Synchronization	Brittle SOAP APIs, no two-way sync, requires manual re-keying of orders and milestones.	Modern RESTful JSON API with bi-directional synchronization, external ID routing (<code>SAP-ORD-xxx</code>), and CDC polling feed.
Network & Service Resilience	Failed API calls are lost or crash batch imports; no retry queue.	Progressive Backoff Ladder (<code>0s → 1m → 5m → 15m → 1h → Dead-Letter</code>) with Admin Portal manual recovery controls.
Builder / Lot Hierarchy	Flat job lists or clumsy parent-child tagging; cannot model national homebuilder subdivisions.	Native 3-Tier Hierarchy: Builder Account → Community / Subdivision → Lot / Unit Phase with inherited attributes.
Field Tech Mobility	Clunky desktop web pages requiring constant cellular connectivity; unusable in basement/concrete dead zones.	Offline-First PWA: Full offline caching via IndexedDB & Service Workers; automatic sync upon reconnection.
Shop Floor Visibility	Paper cut tickets and physical job folders subject to loss, coffee stains, and fabrication errors.	Touchscreen Kiosk View: High-contrast, glove-friendly station barcodes, instant cut list inspection, and real-time station advancing.
Remnant & Slab Utilization	High scrap rates, lost remnant slabs, and unverified inventory counts.	Granular Slab Lifecycle: Full remnant tracking (<code>Original → In-Cut → Remnant → Consumed</code>), QR bin labels, and lot allocations.
Flexible Data Collection	Rigid fixed schema requiring expensive vendor customizations.	Visual Form Matrix Builder & Dynamic Custom Fields supporting multi-row tabular matrices, signature pads, and photo capture.

3. Core Architectural Modules

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    ERP[SAP S/4HANA / Enterprise ERP] <-->|REST API / External IDs| API[SlabMaster REST API & CDC Engine]
    API <--> CORE[Core Multi-Tenant Backend Fastify + Prisma + PostgreSQL]
    CORE <--> QUEUE[Outbound Retry Queue & Dead-Letter Service]
    CORE <--> APP[SlabMaster Web Application React 18 + Vite]
    APP <--> MOD1[Account, Community & Lot Hierarchy]
    APP <--> MOD2[Auto-Schedule & Multi-Plant Calendar]
    APP <--> MOD3[Slab & Remnant Inventory Management]
    APP <--> MOD4[Purchasing & Vendor PO Management]
    APP <--> MOD5[Table Matrix Forms & Custom Attributes]
    APP <--> MOD6[Shop Floor Touchscreen Kiosk]
    APP <--> MOD7[Field Dispatch & Subcontractor PWA]
    APP <--> MOD8[Admin Portal & API Token Management]
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Module Breakdown

- **Tier 1: Enterprise Integration:** High-throughput REST API, SAP Developer Documentation Pack, and downloadable Postman v2.1 collection.
- **Tier 2: Manufacturing & Logistics:** Shop floor station trackers, sawjet cut lists, barcode scanners, and inventory replenishment.
- **Tier 3: Commercial & Residential Scheduling:** 14-day visual matrix, lead-time offsets, plant operating hours, and crew balancing.
- **Tier 4: Field Operations:** Mobile PWA, digital customer sign-offs, photo attachments, and offline form execution.
- **Tier 5: Platform Governance:** Microsoft Entra ID SSO, multi-plant scoping, granular RBAC, and full audit trails.

4. Key Business Value & Strategic ROI

1. **Elimination of Dry-Run Trips (Est. Savings: \$450 - \$750 per avoided trip):** Automated readiness checks and real-time builder milestone alignment prevent installation crews from arriving at unready job sites.
2. **Zero Double-Entry Overhead:** Direct bi-directional synchronization with SAP/ERP eliminates manual order entry, reducing clerical overhead by 80% and removing human transcription errors.
3. **Optimized Stone Yield & Remnant Monetization:** Systematic remnant tracking and bin location tagging allow fabricators to repurpose offcuts for vanities and powder rooms instead of discarding them as scrap.

4. **Rapid Contractor & Crew Onboarding:** Self-service Microsoft Entra ID invitations and scoped 1099 contractor portals allow third-party installers to be onboarded in minutes without risking proprietary pricing leaks.
 5. **Real-Time Job Profitability & Traceability:** Every slab, purchase order, cut ticket, and field modification is permanently linked to the specific Lot and Job, ensuring full financial traceability.
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5. Summary of System Maturity & Quality Metrics

- **Application Version:** `v1.0.0` (production-ready).
 - **Automated Test Coverage:** 39 tests across 9 comprehensive test files covering inventory, purchasing, custom forms, shop floor, table matrices, help documentation, and API retry queues (100% pass rate).
 - **Build Quality:** Strict TypeScript typing (`tsc`) with zero errors across frontend and backend.
 - **CI/CD Integration:** Azure Static Web Apps native deployment with automated build-time test gates.
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